Reverse-Engineering Competition: Inter-Team Evaluation

Objective

Each team will evaluate the reverse-engineering work of all other teams.

Your goal is to assess how well each team's implementation explains and reconstructs the original `spike_data.npz` recording.

Important: We only have the recorded signal, not the original hidden components. Therefore, this is an approximation.

Remember that the random seed used for the original simulation was 42.

Part 1 — Standardize your submission

Each team must create one output file:

`team_<ID>_reconstruction.npz`

It must contain:

Variable	Shape	Description
`t`	`(10000,)`	Time vector
`drift`	`(10000,)`	Estimated drift
`spike_signal`	`(10000,)`	Estimated spike waveform train
`noise`	`(10000,)`	Generated noise
`reconstruction`	`(10000,)`	Sum of all three components
`spike_indices`	`(n_spikes,)`	Estimated spike starting indices

The reconstruction should contain the estimated structured signal (drift + spikes), **not a newly generated noise realization**.

Also submit a short `README.md` describing:

- Your inferred model and parameters.
- Your assumptions.
- Your detection and reconstruction method.

Part 2 — Create an evaluator

Create a Python file called:

`evaluate_team.py`

It must implement the following interface:

```
def evaluate(reference_file, candidate_file):
    """
    Inputs:
        reference_file: path to original spike_data.npz
        candidate_file: path to another team's reconstruction

    Output:
        dictionary containing:
        - validity checks - opt
        - numerical evaluation metrics - opt
        - Overlay figure - opt
        - final score - Must
    """
```

Make sure your code works

To make it more interesting, we will do the actual scoring tomorrow in class :)

Part 3 — Tomorrow in class

Run your evaluator on **every team's submission**.

Including your own, we want to see if there’s bias

For every evaluation, save a result file

and write the team’s scores in the table.